

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(r'C:\Users\hp\Downloads\banking_data.csv')
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
print(df.head())
print(df.shape)
print(df.columns)

print("Missing Values:\n")
print(df.isnull().sum())

#Very few missing values

#Minor missing entries found in: Marital, Marital_status and
Education

#marital and marital_status represent the same information
df.drop(columns = ['marital'], inplace=True)
print(df.columns)
```

#day_month is derived field from day and month, both of which are already available separately and provide clearer interpretability.

```
df.drop(columns = ['day_month'], inplace=True)
print(df.columns)
```

```
unique_counts = df.nunique()
print(unique_counts)
```

```
plt.hist(df['age'], bins=30, edgecolor='black')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Age distribution')
plt.show()
```

```
job_counts = df['job'].value_counts()
print(job_counts)
print((df['job'].value_counts(normalize=True) * 100).round(1))
sns.countplot(y='job', data=df, order=job_counts.index)
plt.title("Job Type Distribution")
plt.xlabel("Count")
plt.ylabel("Job")
plt.show()
```

```
marital_status_counts= df['marital_status'].value_counts()
print(marital_status_counts)
print((df['marital_status'].value_counts(normalize=True) *
100).round(1))

sns.countplot(x='marital_status', data=df,
order=marital_status_counts.index)

plt.title("Marital Status Distribution")
plt.xlabel("Marital Status")
plt.ylabel("Count")
plt.show()
```

```
education_value_counts= df['education'].value_counts()
print(education_value_counts)
print((df['education'].value_counts(normalize=True) * 100).round(1))
order = ['primary', 'secondary', 'tertiary', 'unknown', 'nan']
education_counts = education_value_counts.reindex(order)
sns.countplot(x='education', data=df, order=education_counts.index)
plt.title("Education Level Distribution")
plt.xlabel("Education")
plt.ylabel("Count")
plt.show()
```

```
credit_in_default_counts = df['default'].value_counts()
print(credit_in_default_counts)
print((df['default'].value_counts(normalize=True) * 100).round(1))
sns.countplot(x='default', data=df,
order=credit_in_default_counts.index)
plt.title("Credit Default Distribution")
plt.xlabel("Credit in Default")
plt.ylabel("Count")
plt.show()
```

```
print(df['balance'].describe())
plt.figure(figsize=(8,5))
sns.kdeplot(df['balance'], fill=True, linewidth=2)
plt.title("Density Curve of Average Yearly Balance")
plt.xlabel("Average Yearly Balance")
plt.show()
```

```
housing_loan_counts = df['housing'].value_counts()
print(housing_loan_counts)
print((df['housing'].value_counts(normalize=True) * 100).round(1))
```

```
sns.countplot(x='housing', data=df,  
order=housing_loan_counts.index)  
plt.title("Housing Loan Distribution")  
plt.xlabel("Housing Loan")  
plt.ylabel("Count")  
plt.show()
```

```
loan_counts = df['loan'].value_counts()  
print(loan_counts)  
print((df['loan'].value_counts(normalize=True) * 100).round(1))  
sns.countplot(x='loan', data=df, order=loan_counts.index)  
plt.title("Personal Loan Distribution")  
plt.xlabel("Personal Loan")  
# plt.ylabel("Count")  
# plt.show()
```

```
contact_counts = df['contact'].value_counts()  
print(contact_counts)  
print((df['contact'].value_counts(normalize=True) * 100).round(1))  
sns.countplot(x='contact', data=df, order=contact_counts.index)  
plt.title("Communication Type Distribution")  
plt.xlabel("Contact Type")
```

```
plt.ylabel("Count")  
plt.show()
```

```
print(df['day'].describe())  
sns.histplot(df['day'], bins=31, kde=False)  
plt.title("Distribution of Last Contact Day")  
plt.xlabel("Day of Month")  
plt.ylabel("Count")  
plt.show()
```

```
month_counts = df['month'].value_counts()  
print(month_counts)  
print((df['month'].value_counts(normalize=True) * 100).round(1))  
month_order =  
['jan', 'feb', 'mar', 'apr', 'may', 'jun', 'jul', 'aug', 'sep', 'oct', 'nov', 'dec']  
sns.countplot(x='month', data=df, order=month_order)  
plt.title("Distribution of Last Contact Month")  
plt.xlabel("Month")  
plt.ylabel("Count")  
plt.show()
```

```
print(df['duration'].describe())  
plt.hist(df['duration'], bins=50, edgecolor='black')  
plt.title("Distribution of Call Duration")
```

```
plt.xlabel("Duration (seconds)")  
plt.ylabel("Count")  
plt.show()
```

```
print(df['campaign'].describe())  
plt.hist(df['campaign'], bins=30, edgecolor='black')  
plt.title("Distribution of Number of Contacts During Campaign")  
plt.xlabel("Campaign Contacts")  
plt.ylabel("Count")  
plt.show()
```

```
print(df['pdays'].describe())  
plt.hist(df['pdays'], bins=50, edgecolor='black')  
plt.title("Distribution of Days Since Previous Contact")  
plt.xlabel("pdays")  
plt.ylabel("Count")  
plt.show()
```

```
print(df['previous'].describe())  
plt.hist(df['previous'], bins=30, edgecolor='black')  
plt.title("Distribution of Previous Contacts")  
plt.xlabel("Previous Contacts")  
plt.ylabel("Count")  
plt.show()
```

```
poutcome_counts = df['poutcome'].value_counts()
print(poutcome_counts)
print((df['poutcome'].value_counts(normalize=True) * 100).round(1))
sns.countplot(x='poutcome', data=df, order=poutcome_counts.index)
plt.title("Outcome of Previous Marketing Campaign")
plt.xlabel("Previous Outcome")
plt.ylabel("Count")
plt.show()
```

```
y_counts = df['y'].value_counts()
print(y_counts)
sns.countplot(x='y', data=df, order=y_counts.index)
plt.title("Term Deposit Subscription Distribution")
plt.xlabel("Subscribed?")
plt.ylabel("Count")
plt.show()
print((df['y'].value_counts(normalize=True) * 100).round(2))
```

```
numeric_cols = ['age', 'balance', 'day', 'duration', 'campaign', 'pdays',
'previous', 'y_numeric']
corr = df[numeric_cols].corr()
```

```
print(corr['y_numeric'].sort_values(ascending=False))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

```
df['y_numeric'] = df['y'].map({'no': 0, 'yes': 1})
df[['y', 'y_numeric']].head()
```

```
job_sub = pd.crosstab(df['job'], df['y'], normalize='index') * 100
print(job_sub)
job_sub['yes'].sort_values().plot(kind='barh')
plt.title("Subscription Rate by Job")
plt.xlabel("Subscription Rate (%)")
plt.ylabel("Job")
plt.show()
```

```
marital_sub = pd.crosstab(df['marital_status'], df['y'],
normalize='index') * 100
print(marital_sub)
marital_sub['yes'].plot(kind='bar')
plt.title("Subscription Rate by Marital Status")
plt.ylabel("Subscription Rate (%)")
plt.xlabel('Marital status')
```

```
plt.show()
```

```
education_sub = pd.crosstab(df['education'], df['y'],  
normalize='index') * 100
```

```
print(education_sub)
```

```
education_sub['yes'].plot(kind='bar')
```

```
plt.title("Subscription Rate by Education")
```

```
plt.ylabel("Subscription Rate (%)")
```

```
plt.xlabel('Education level')
```

```
plt.show()
```

```
default_sub = pd.crosstab(df['default'], df['y'], normalize='index') *  
100
```

```
print(default_sub)
```

```
default_sub['yes'].plot(kind='bar')
```

```
plt.title("Subscription Rate by Default")
```

```
plt.ylabel("Subscription Rate (%)")
```

```
plt.xlabel('Credit in Default')
```

```
plt.show()
```

```
housing_sub = pd.crosstab(df['housing'], df['y'], normalize='index') *  
100
```

```
print(housing_sub)
```

```
housing_sub['yes'].plot(kind='bar')
```

```
plt.title("Subscription Rate by Housing Loan")
```

```
plt.ylabel("Subscription Rate (%)")
plt.xlabel('Housing Loan')
plt.show()
```

```
loan_sub = pd.crosstab(df['loan'], df['y'], normalize='index') * 100
print(loan_sub)
loan_sub['yes'].plot(kind='bar')
plt.title("Subscription Rate by Personal Loan")
plt.ylabel("Subscription Rate (%)")
plt.xlabel('Personal Loan')
plt.show()
```

```
contact_sub = pd.crosstab(df['contact'], df['y'], normalize='index') *
100
print(contact_sub)
contact_sub['yes'].plot(kind='bar')
plt.title("Subscription Rate by Contact Type")
plt.ylabel("Subscription Rate (%)")
plt.xlabel('Contact type')
plt.show()
```

```
month_sub = pd.crosstab(df['month'], df['y'], normalize='index') *
100
print(month_sub)
```

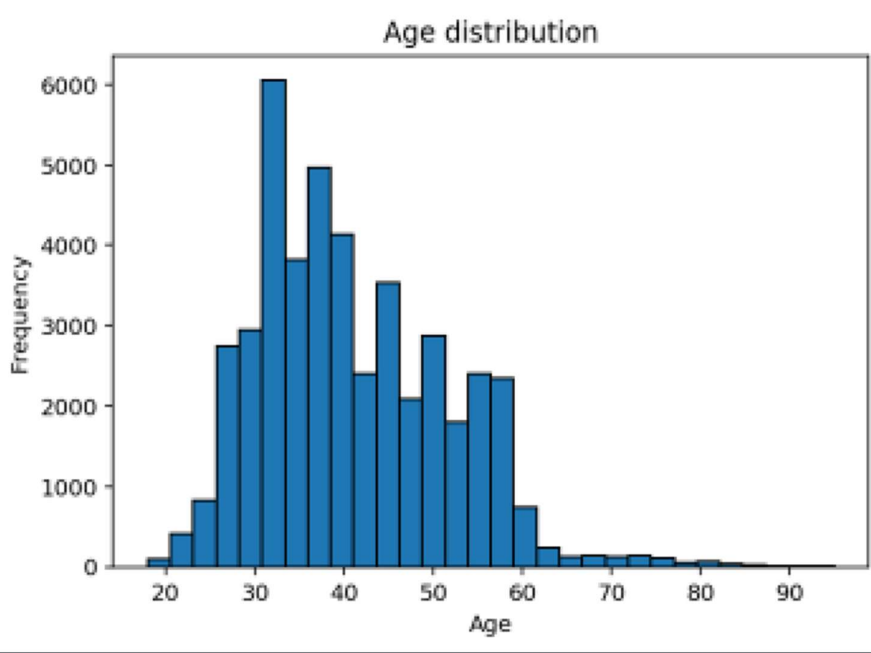
```
month_order =  
['jan','feb','mar','apr','may','jun','jul','aug','sep','oct','nov','dec']  
month_sub = month_sub.reindex(month_order)  
month_sub['yes'].plot(kind='bar')  
plt.title("Subscription Rate by Month")  
plt.ylabel("Subscription Rate (%)")  
plt.show()
```

```
poutcome_sub = pd.crosstab(df['poutcome'], df['y'],  
normalize='index') * 100  
print(poutcome_sub)  
poutcome_sub['yes'].plot(kind='bar')  
plt.title("Subscription Rate by Previous Campaign Outcome")  
plt.ylabel("Subscription Rate (%)")  
plt.show()
```

#In all of the countplots, Count = Number of clients.

Solutions of case questions:

Q1. What is the distribution of age among the clients?



Findings:

- **Minimum age: 18**
- **Maximum age: 95**
- **Average age: 40.94 years**
- **Median age: 39 years**
- **Most customers are concentrated in the 30–50 age group**

Interpretation:

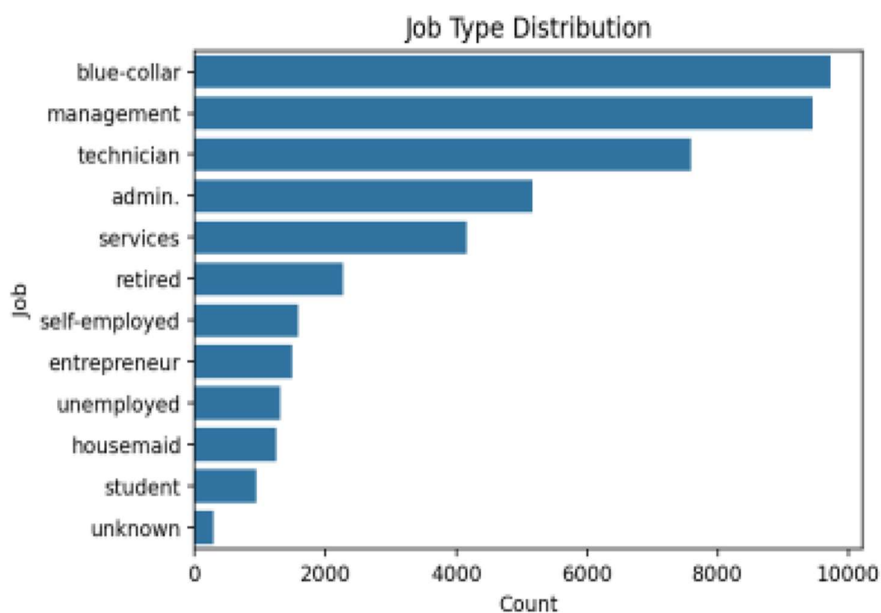
The client base is primarily made up of **working-age adults**, which is expected for banking and deposit products.

Business Insight:

Term deposit campaigns may perform best when tailored to:

- **middle-aged customers** with stable income
- **older customers** nearing retirement who prefer safer investments

Q2. How does the job type vary among the clients?



Most common job categories:

- **Blue-collar:** 21.5%
- **Management:** 20.9%
- **Technician:** 16.8%
- **Admin:** 11.4%
- **Services:** 9.2%

Less common categories:

- Student: 2.1%
- Retired: 5.0%
- Unemployed: 2.9%
- Housemaid: 2.7%
- Unknown: 0.6%

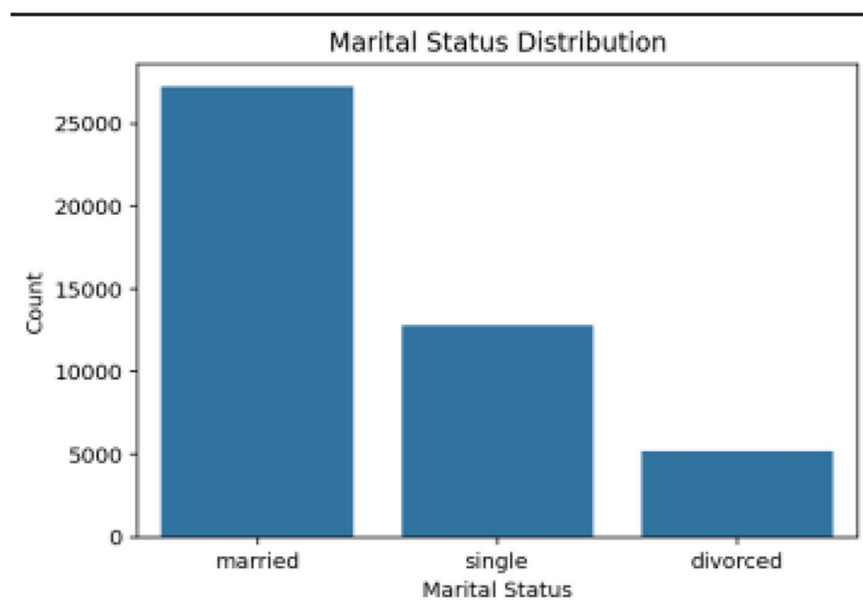
Interpretation:

The customer base is heavily concentrated in **working and salaried occupations**, especially blue-collar, management, and technical roles.

Business Insight:

This indicates the bank serves a broad mass-market and working professional segment.

Q3. What is the marital status distribution of the clients?



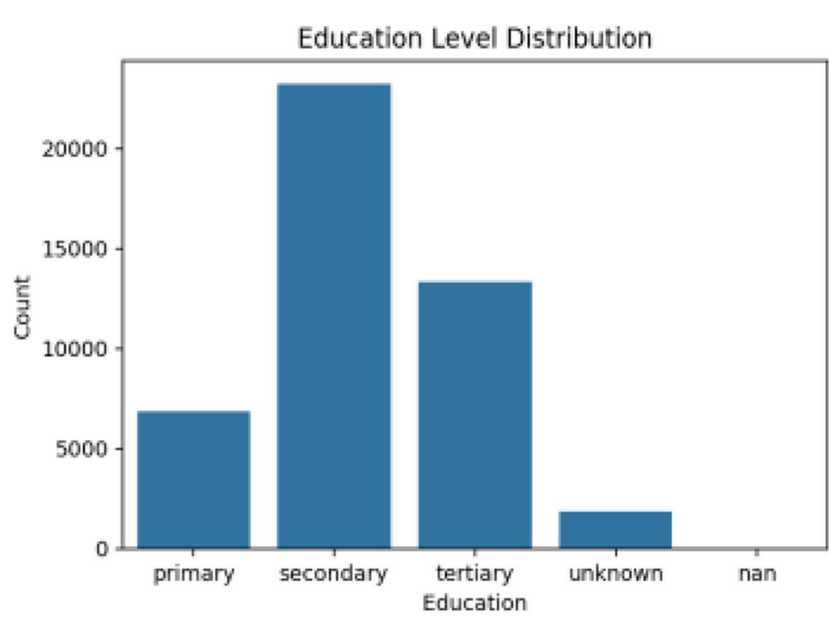
Distribution:

- **Married:** 60.2%
- **Single:** 28.3%
- **Divorced:** 11.5%

Interpretation:

A majority of the bank's customers are **married**, which may reflect more stable financial behavior and longer-term savings patterns.

Q4. What is the level of education among the clients?



Distribution:

- **Primary education:** 15.2%
- **Secondary education:** 51.3%
- **Tertiary education:** 29.4%

- **Unknown: 4.1%**

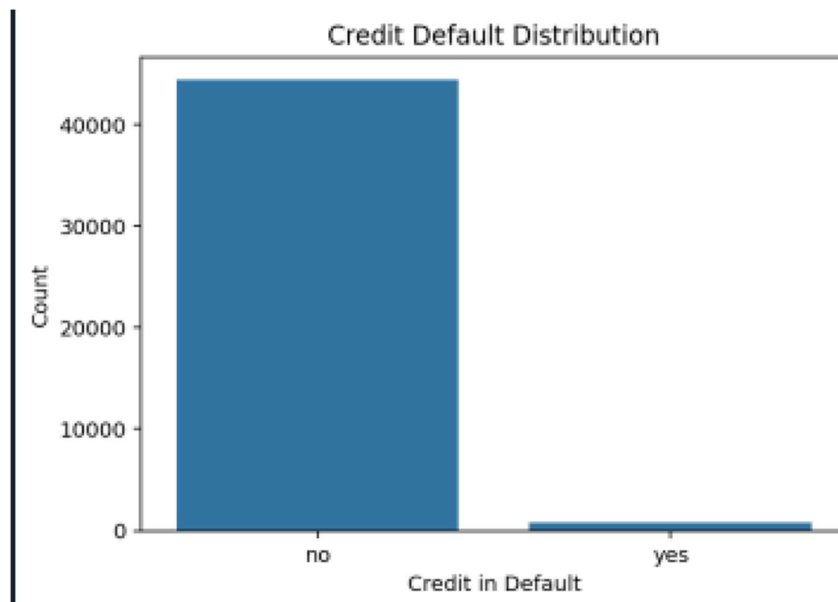
Interpretation:

Most clients have at least **secondary education**, with a substantial proportion having **tertiary education**.

Business Insight:

Customers may respond well to **clear, product-benefit-based communication**, especially those with higher education who may better understand financial products.

Q5. What proportion of clients have credit in default?



Distribution:

- **No default: 98.2%**
- **Default: 1.8%**

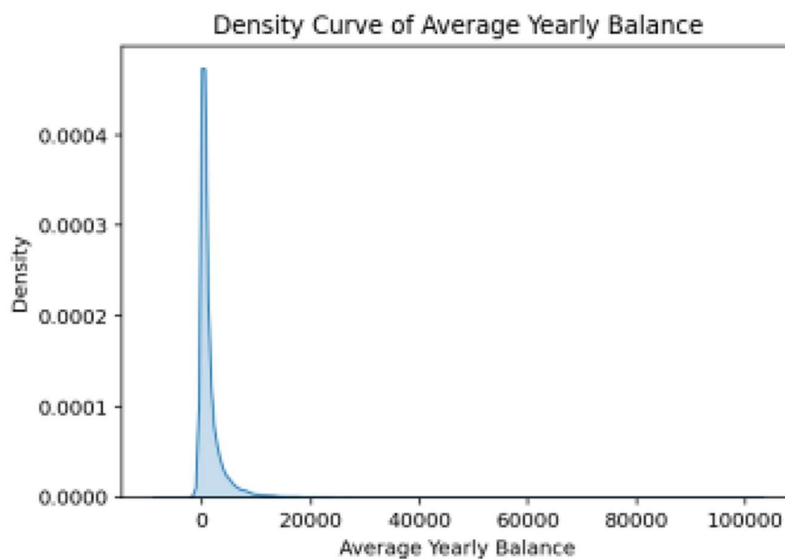
Interpretation:

Only a very small fraction of customers are in credit default.

Business Insight:

This is a relatively healthy customer base from a risk perspective.

Q6. What is the distribution of average yearly balance among the clients?



Summary:

- **Average balance:** €1,362
- **Median balance:** €448.5
- **Minimum:** -€8,019
- **Maximum:** €102,127

Interpretation:

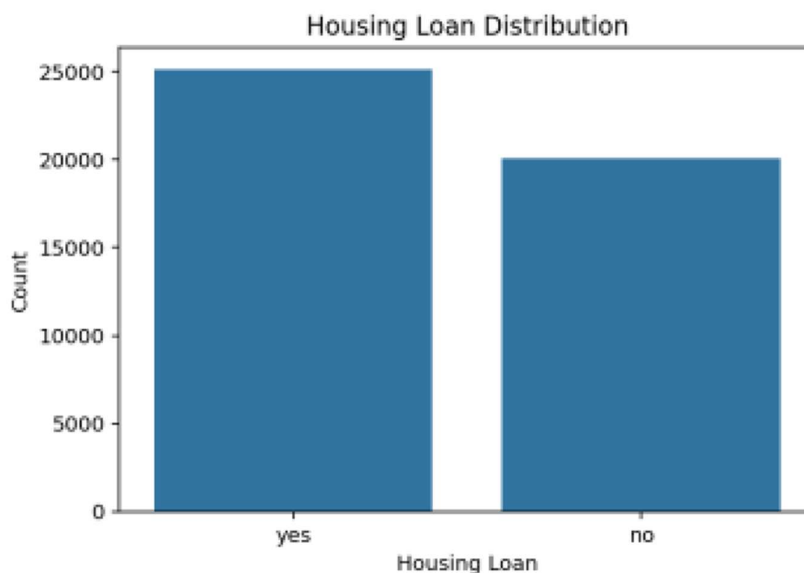
Balances are **highly skewed**:

- many customers have **low or moderate balances**
- a few customers have **very high balances**

Business Insight:

Customers with **higher balances** may be more suitable for term deposit targeting, as they likely have surplus funds to invest.

Q7. How many clients have housing loans?



Distribution:

- **Yes:** 55.6%
- **No:** 44.4%

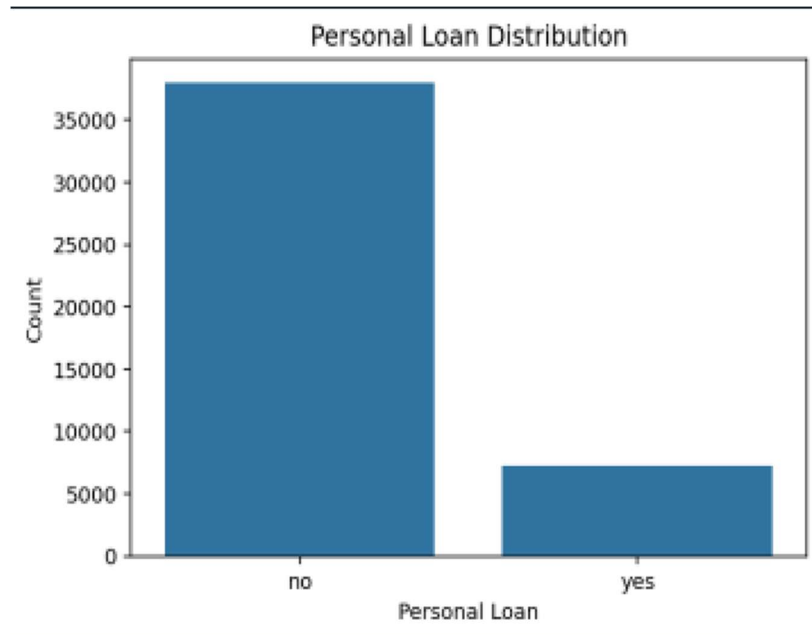
Interpretation:

A majority of customers have a housing loan.

Business Insight:

Customers with housing loans may have **less spare liquidity**, which can influence term deposit conversion.

Q8. How many clients have personal loans?



Distribution:

- **No:** 84.0%
- **Yes:** 16.0%

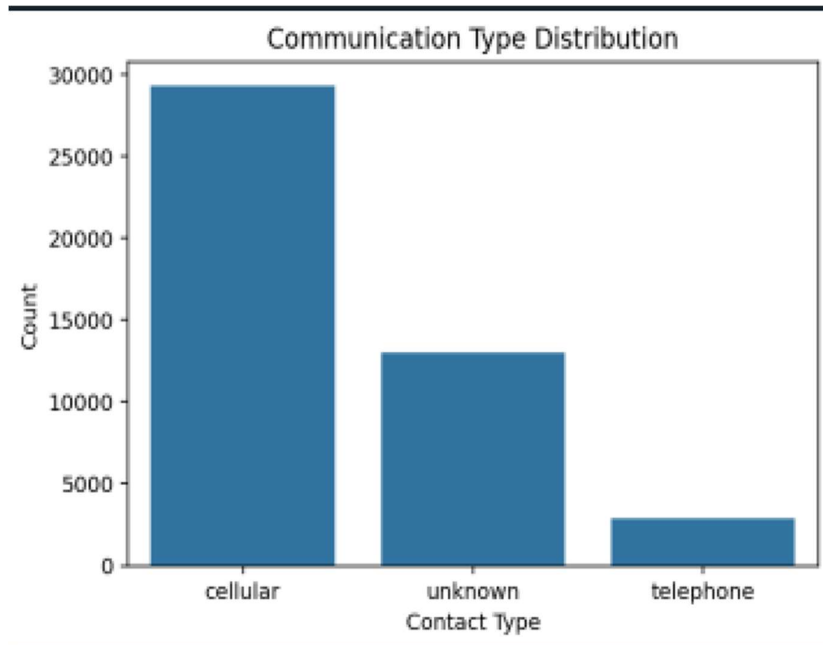
Interpretation:

Only a smaller portion of clients have personal loans.

Business Insight:

Customers without personal loans may have greater financial flexibility for deposits.

Q9. What are the communication types used for contacting clients during the campaign?



Distribution:

- **Cellular:** 64.8%
- **Unknown:** 28.8%
- **Telephone:** 6.4%

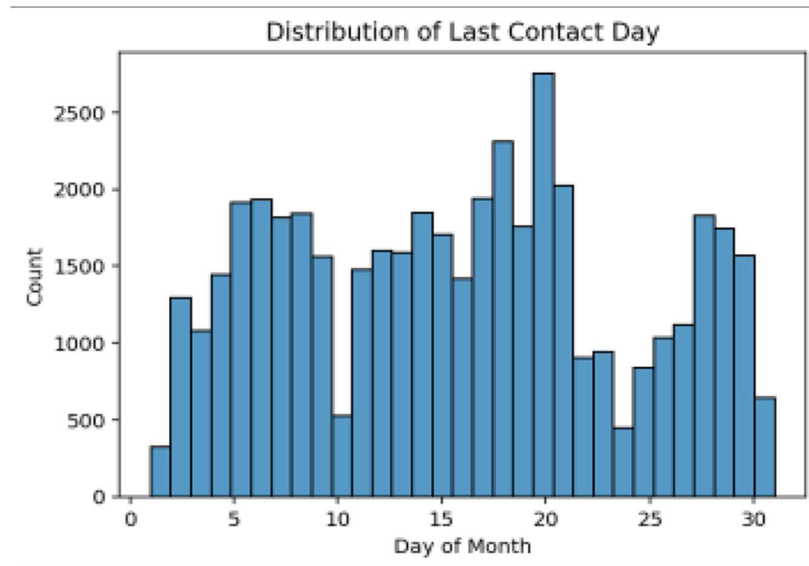
Interpretation:

Most customers were contacted through **cellular/mobile channels**.

Business Insight:

Mobile outreach appears to be the dominant communication channel and may be the most effective one to optimize.

Q10. What is the distribution of the last contact day of the month?



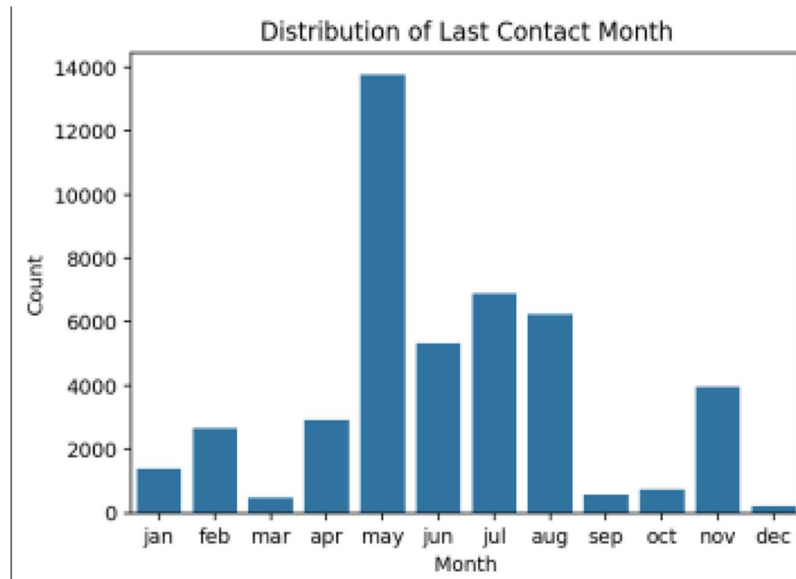
Observation:

The day variable ranges across the days of the month, but by itself it is **less strategically meaningful** than broader time-based variables such as month.

Interpretation:

No strong business conclusion can be drawn from day-of-month alone without more detailed time-series analysis.

Q11. How does the last contact month vary among the clients?



Most active campaign months:

- **May:** 30.4%
- **July:** 15.2%
- **August:** 13.8%
- **June:** 11.8%
- **November:** 8.8%

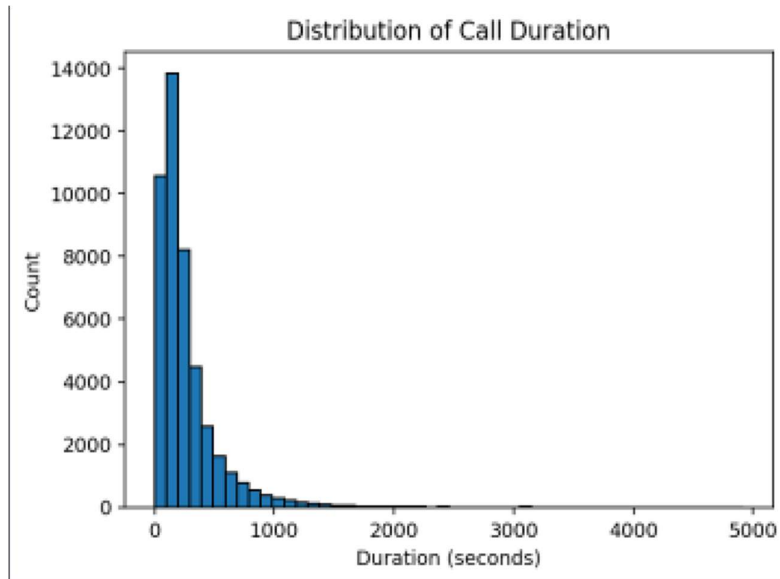
Interpretation:

Campaign activity was concentrated in a few specific months, especially **May**.

Business Insight:

Campaign timing is not evenly distributed, which may affect conversion outcomes.

Q12. What is the distribution of the duration of the last contact?



Summary:

- **Average duration:** 258 seconds
- **Median duration:** 180 seconds
- **Minimum:** 0 seconds
- **Maximum:** 4,918 seconds

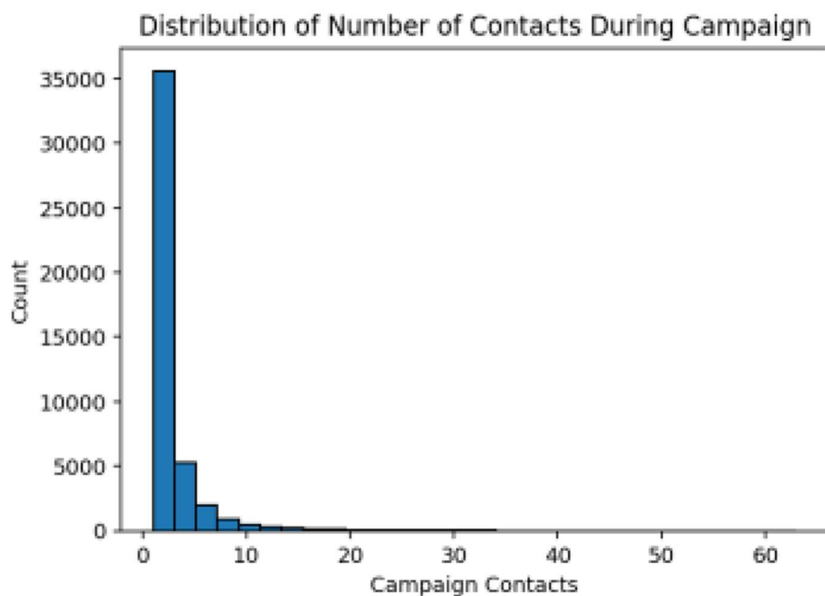
Interpretation:

Call durations vary widely, with some very short calls and some very long ones.

Business Insight:

Longer conversations often indicate **higher customer engagement**.

Q13. How many contacts were performed during the campaign for each client?



Summary:

- **Average contacts: 2.76**
- **Median contacts: 2**
- **Maximum: 63**

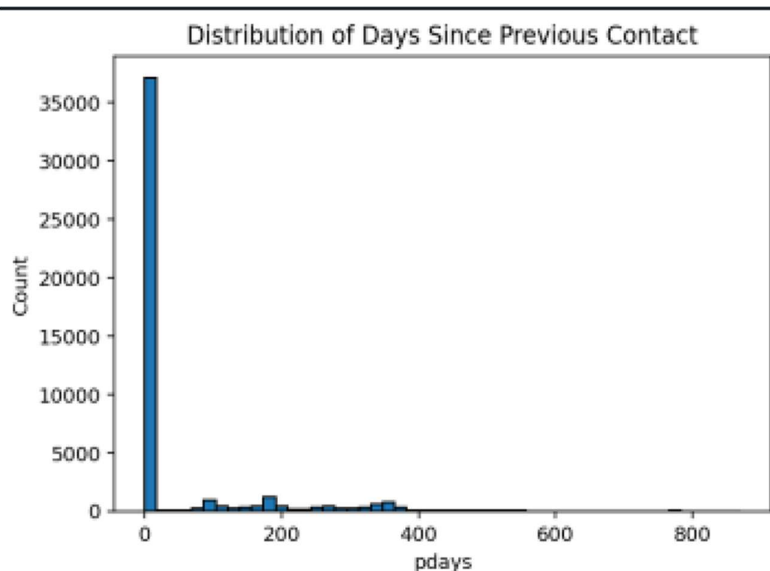
Interpretation:

Most customers were contacted only a **few times**, but some required repeated attempts.

Business Insight:

Too many repeated contacts may indicate **inefficient targeting** or lower interest.

Q14. What is the distribution of the number of days passed since the client was last contacted from a previous campaign?



Summary:

- pdays = -1 for most customers

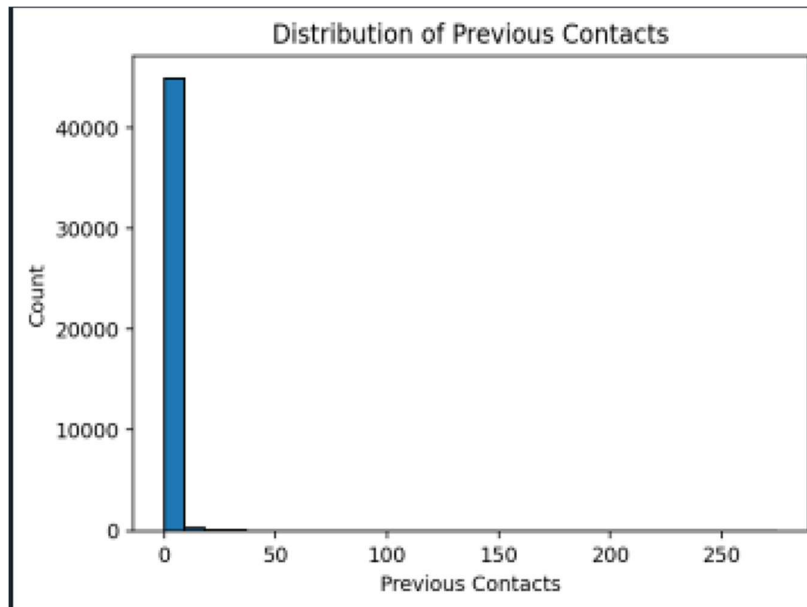
Interpretation:

A value of **-1** means the customer was **not contacted previously**, so most customers had **no recent prior campaign contact**.

Business Insight:

This suggests a large portion of the campaign may involve **fresh outreach rather than repeated re-engagement**.

Q15. How many contacts were performed before the current campaign for each client?



Summary:

- **Average previous contacts:** 0.58
- **Median:** 0
- **Maximum:** 275

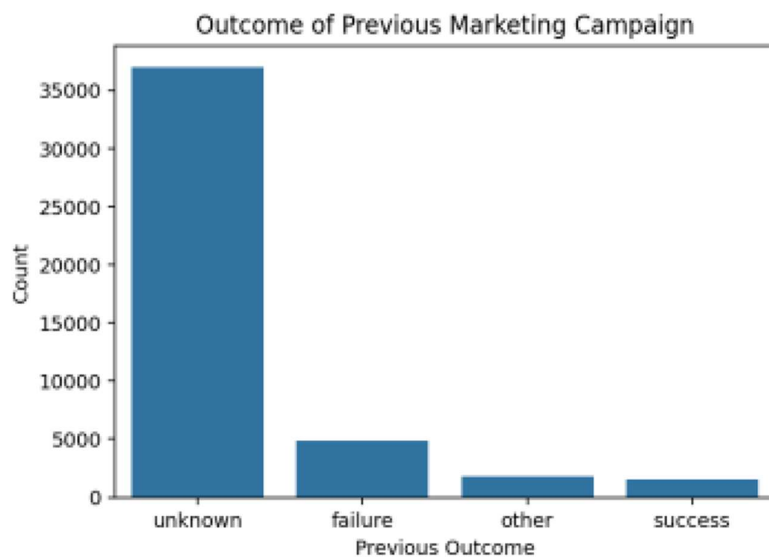
Interpretation:

Most customers had **no previous contacts**, while a small number were contacted many times before.

Business Insight:

Repeated historical contact may help indicate prior engagement, but too much repetition may also reflect low conversion efficiency.

Q16. What were the outcomes of the previous marketing campaigns?



Observation:

Most customers fall under:

- **Unknown: 81.7%**
- **Failure: 10.8%**
- **Other: 4.1%**
- smaller portion in **success: 3.1%**

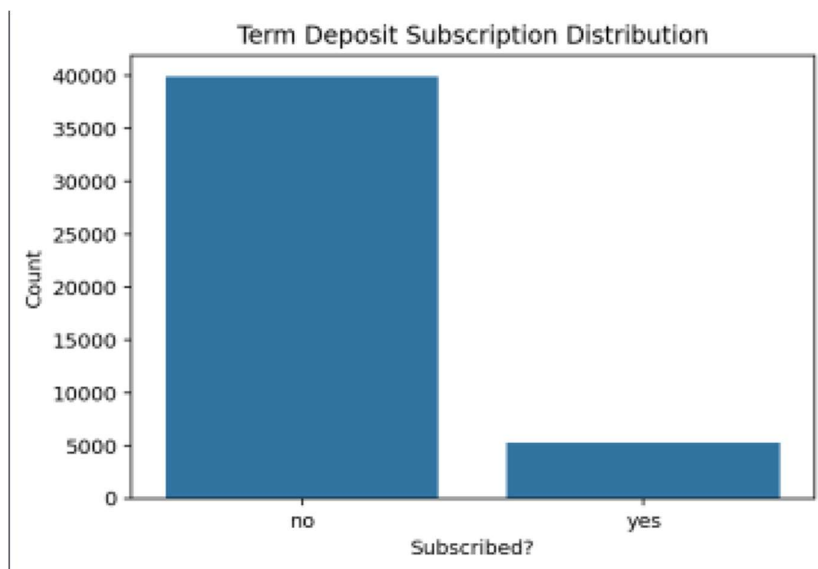
Interpretation:

Previous campaign outcome is very important because it reflects the customer's past response to similar outreach.

Business Insight:

Customers who previously responded positively are much more valuable leads.

Q17. What is the distribution of clients who subscribed to a term deposit vs. those who did not?



Distribution:

- **Subscribed (yes): 11.7%**
- **Did not subscribe (no): 88.3%**

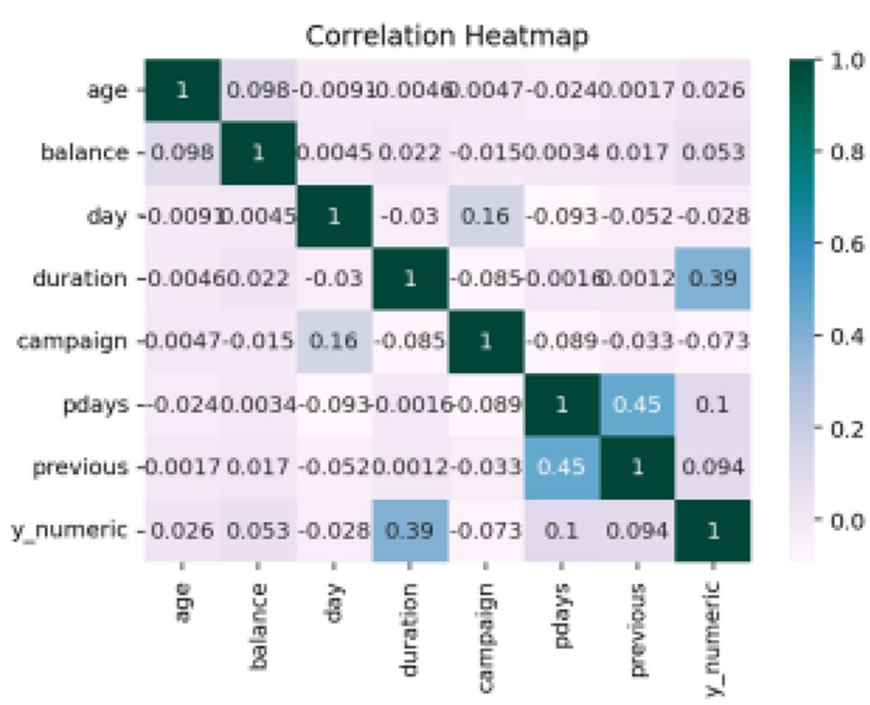
Interpretation:

The vast majority of customers did **not** subscribe.

Business Insight:

This means the bank should **not call everyone equally**.
Instead, it should identify **high-probability customers** and prioritize them.

Q18. Are there any correlations between different attributes and the likelihood of subscribing to a term deposit?



A) Numeric variables most related to subscription

Strongest positive relationships:

- **Duration: +0.394**
- **Pdays: +0.104**
- **Previous: +0.094**
- **Balance: +0.053**
- **Age: +0.026**

Negative relationships:

- Campaign: -0.073
- Day: -0.028

Interpretation:

1. Duration is the strongest predictor

Longer calls are associated with higher subscription likelihood.

BUT: this should not be used for pre-campaign targeting because you only know duration **after the call happens**.

2. Campaign has a negative relationship

More repeated calls during the same campaign are associated with **lower probability of subscription**.

Business implication

If a customer is not converting after multiple calls, the bank should consider **stopping or deprioritizing further calls**.

3. Previous engagement matters

Customers with:

- more previous contacts
 - shorter or meaningful prior outreach history
- are slightly more likely to convert.

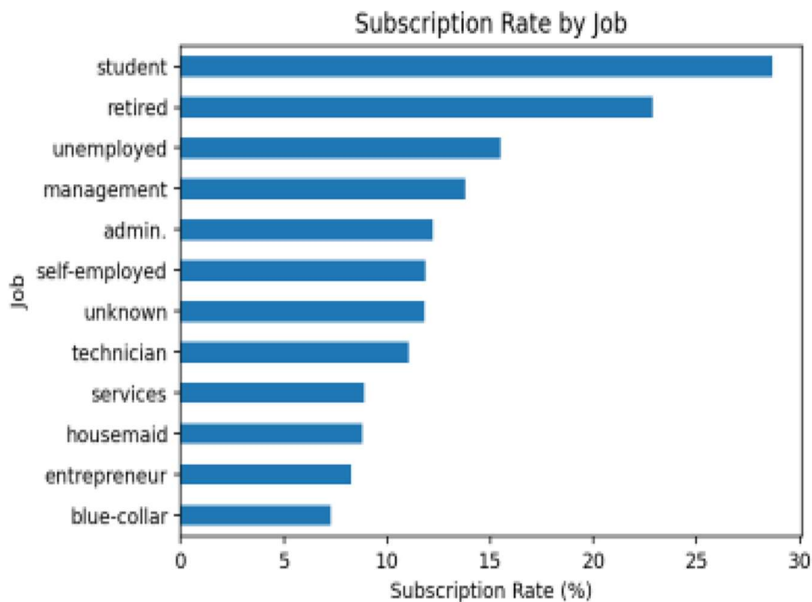
Business implication:

Past engagement history can be useful in identifying warmer leads.

6) Which Customer Groups Convert Better?

These are especially useful for targeting.

A) Job-wise subscription rate



Highest converting job groups:

- **Student:** 28.7%
- **Retired:** 22.9%
- **Unemployed:** 15.5%
- **Management:** 13.8%
- **Admin:** 12.2%

Interpretation:

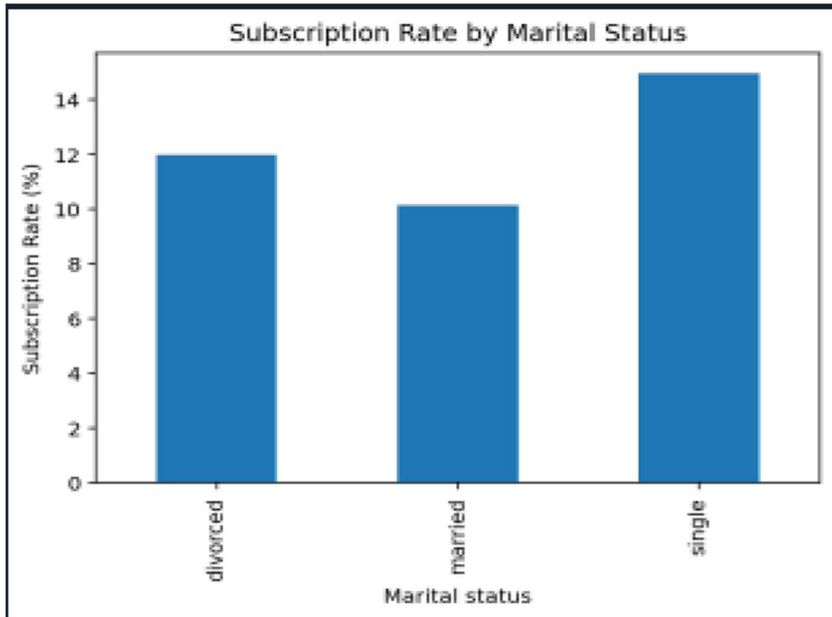
Students and retired individuals show surprisingly strong conversion rates.

Business Insight:

These groups may be:

- more responsive to savings products
- more interested in safe fixed-return instruments

B) Marital status vs subscription



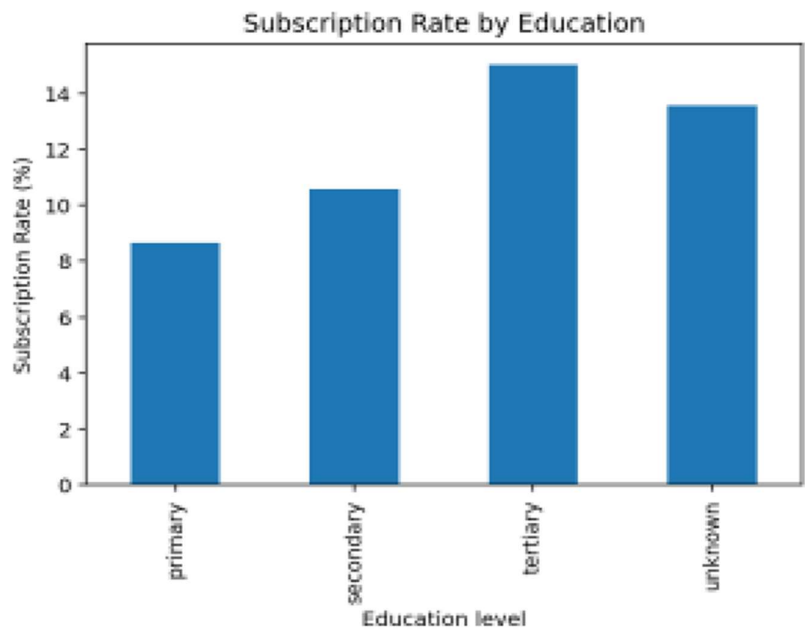
Subscription rate:

- **Single:** 15.0%
- **Divorced:** 12.0%
- **Married:** 10.1%

Interpretation:

Single customers are more likely to subscribe than married customers.

C) Education vs subscription



Subscription rate:

- **Tertiary:** 15.0%
- **Unknown:** 13.6%
- **Secondary:** 10.6%
- **Primary:** 8.6%

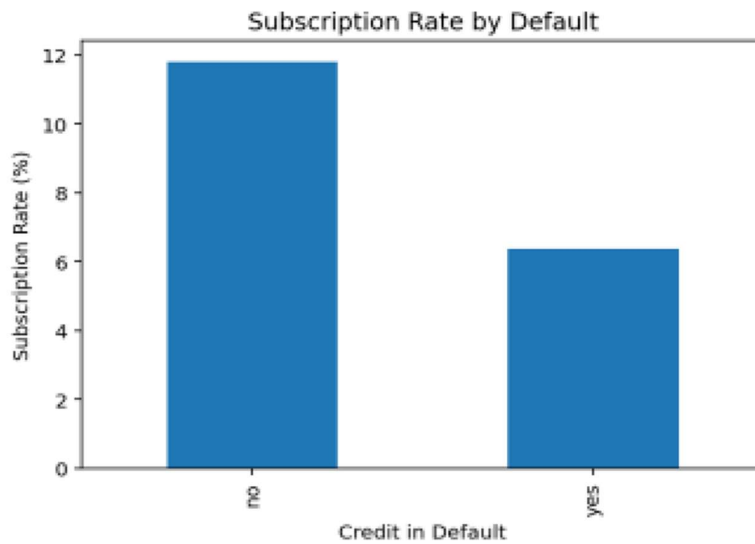
Interpretation:

Higher education is associated with better conversion.

Business Insight

Educated customers may better understand the value of a term deposit.

D) Default vs subscription



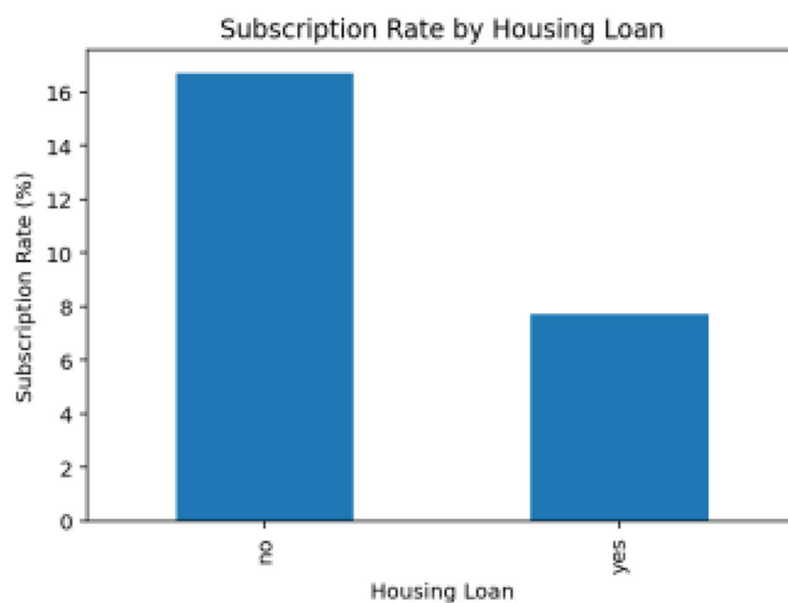
Subscription rate:

- **No default:** 11.8%
- **Default:** 6.4%

Interpretation:

Customers without default history are more likely to subscribe.

E) Housing loan vs subscription



Subscription rate:

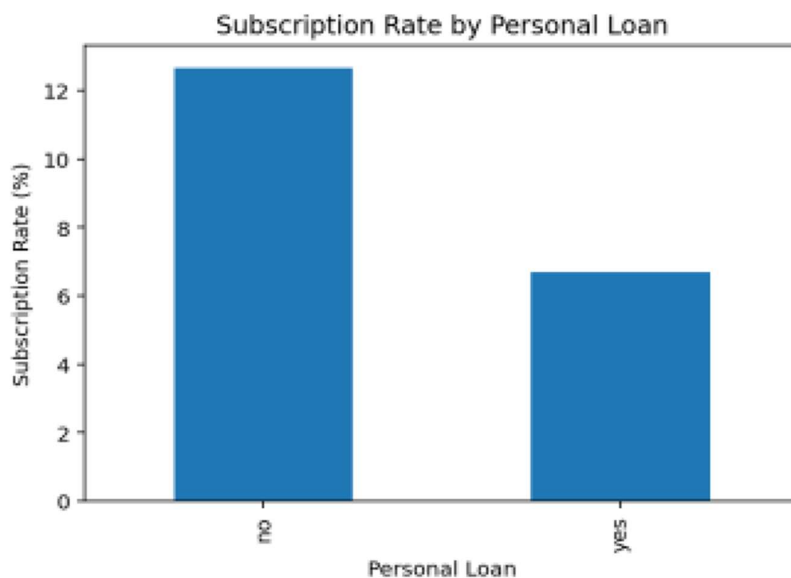
- No housing loan: 16.7%
- Housing loan: 7.7%

Interpretation:

Customers without housing loans are much more likely to subscribe.

Business Insight:

Mortgage burden may reduce disposable savings capacity.

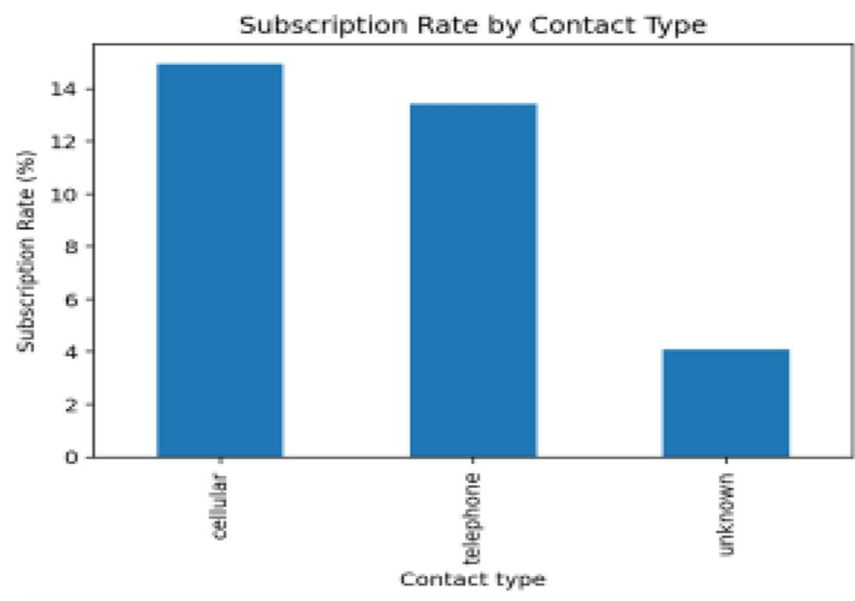
F) Personal loan vs subscription**Subscription rate:**

- No personal loan: 12.7%
- Personal loan: 6.7%

Interpretation:

Customers without personal loans convert better.

G) Contact type vs subscription



Subscription rate:

- **Cellular:** 14.9%
- **Telephone:** 13.4%
- **Unknown:** 4.1%

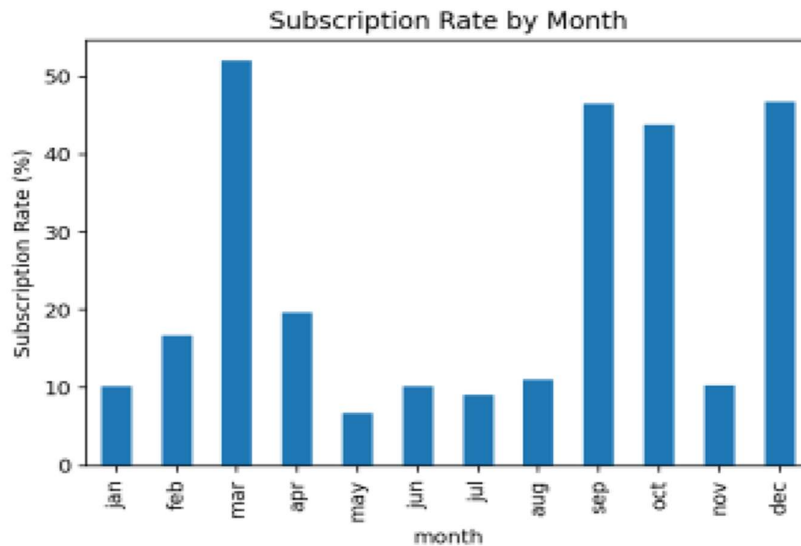
Interpretation:

Customers contacted via **cellular** perform best.

Business Insight:

The bank should prioritize mobile-based outreach where possible.

H) Month vs subscription



Highest subscription months:

- **March:** 52.0%
- **December:** 46.7%
- **September:** 46.5%
- **October:** 43.8%
- **April:** 19.7%

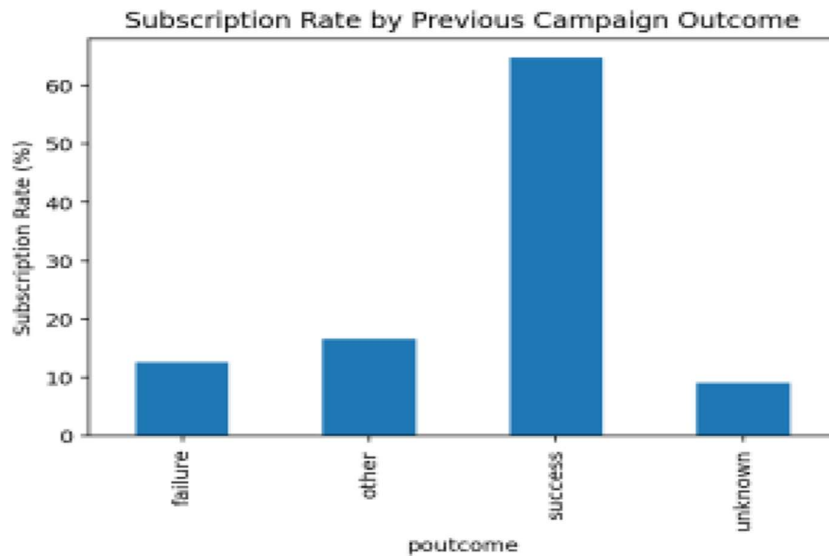
Interpretation:

Some months are much more favorable than others.

Business Insight:

Campaign timing appears to matter significantly.

I) Previous campaign outcome vs subscription



Subscription rate:

- **Success:** 64.8%
- **Other:** 16.7%
- **Failure:** 12.6%
- **Unknown:** 9.2%

Interpretation:

This is one of the **strongest business indicators**.

Business Insight:

Customers who previously responded successfully should be treated as **high-priority leads**.

THANK YOU